

FSK176: Tutorial Problem Numbers / Tutoriaal Probleem Nommers

2015

Chapter 3 Vectors / Hoofstuk 3 Vektore

All the indicated numbers refer to the “Problems” section in Serway; e.g. P1.9 is problem 9 from chapter 1.

Alle nommers verwys na die “Problems” afdeling in Serway; bv. P1.9 is probleem 9 uit hoofstuk 1.

P3.11 (a) 5.2 m at 60° (b) 3.0 m at 330° (c) 3.0 m at 150° (d) 5.2 m at 300°

P3.12 (a) graphical (b) $\vec{R}_1 = \vec{R}_2 = \vec{R}_3$

P3.15 47.2 units at 58.0° above the negative x -axis

P3.20 (a) 5.00 blocks at 53.1° North of East (b) 13.0 blocks

P3.23 (a) $2\hat{i} - 6\hat{j}$ (b) $4\hat{i} + 2\hat{j}$ (c) 6.32 (d) 4.47

P3.45 1.43×10^4 m at 32.2° above the horizontal

P3.51 $(-130 \text{ m})\hat{i} - (202 \text{ m})\hat{j}$ or 240 m at 57.2° below the horizontal

P3.60 $\theta = 2 \tan^{-1} \left(\frac{1}{n} \right)$

P3.61 (a) $|\vec{A} + \vec{B}| = [10000 - (9600) \sin \theta]^{1/2} \text{ cm}$ (b) $\theta = 270^\circ, 140.0 \text{ cm}$

(c) $\theta = 90^\circ, 20.0 \text{ cm}$ (d) explanation

P3.62 $\Delta \vec{r} = (0.456 \text{ m})\hat{i} - (0.708 \text{ m})\hat{j}$

P3.63 (a) $\frac{d\vec{r}}{dt} = -(2.00 \text{ m/s})\hat{k}$ (b) explanation.